

Partner-Specific Affective Precision in Social Active Inference

Anonymous Author(s)

Anonymous Institution

Abstract. Affect in social settings can function as metacognition: not a direct signal of what another agent will do, and not simply a record of reward or punishment, but an estimate of whether the focal agent’s current partner model warrants confident action. This distinction matters in multi-partner environments, where an agent may have a reliable model of one partner, an uncertain model of another, and a predictably hostile model of a third. We formalize this idea in active inference by extending affective precision to relationship-specific model-fitness estimates. Each partner receives its own β estimate, updated only from that partner’s behavioral evidence, which modulates policy precision when acting toward that partner.

Simulations in a multi-partner trust game show that partner-local affective precision tracks predictability rather than value: a reliably defecting and a reliably cooperative partner can both support high precision, because what they share is predictability, not positive value. In decision-open regimes, deployment changes arise through action sharpening rather than improved belief quality; in partner-choice settings, engagement reorganizes around reliability before aggregate payoff separates. Under abrupt behavioral change, relationship-specific confidence reshapes action deployment through policy precision; payoff effects are modest and uncertain, and the temporal benefit of accumulated confidence depends on whether the social portfolio permits rapid reallocation away from the changed partner. Precision gain and priors over model fitness define computational profiles that connect parameterized confidence dynamics to empirically observed variation in social trust. Affect, in this account, is a relationship-specific metacognitive signal whose dynamics reflect the temporal and relational structure of social life.

Keywords: Active inference · Affective precision · Social metacognition · Multi-agent systems · Trust · Individual differences

1 Introduction

Social cognition routinely conflates three problems that are better kept separate. The first is inferring what a partner is like—whether they tend to cooperate, defect, or behave unpredictably. The second is modeling what they know or intend—the theory-of-mind question of how another agent represents the world and plans action [10, 11, 17]. The third is estimating whether one’s *current* partner

model is reliable enough to guide confident action. Accurate type inference and sophisticated mentalizing do not by themselves solve this third problem. Knowing that someone is unreliable is not the same as knowing how cautiously to treat them; knowing that someone is predictable is not the same as knowing how strongly to commit to that prediction. Most formal accounts emphasize the content of social inference while leaving this commitment problem implicit.

We argue that affect in social settings functions as *social metacognition*: not a direct readout of what another agent will do, and not simply a record of reward or punishment, but an estimate of whether the focal agent’s current partner model warrants confident action. In psychology and neuroscience, affect is often described in terms of valence and arousal—the felt pull toward or away from situations, relationships, and social commitments [2]. Functionally, however, it can regulate how strongly beliefs are expressed in action: affect assigns action-relevant weight to what an agent already infers, shaping the degree of commitment rather than the content of inference. An agent that infers accurately but commits poorly wastes its social knowledge; one that commits confidently on fragile models exposes itself to exploitation.

The active inference framework [7, 8, 16] provides a formal route from this observation to a computational mechanism. Agents minimize variational free energy to update beliefs and evaluate candidate policies by their expected free energy [9, 6]. Action selection is precision-weighted: the same beliefs can produce decisive or tentative behavior depending on γ , the parameter controlling how sharply the agent commits to its preferred policies. Hesp et al. [13] made affect endogenous by modeling it as active inference over a higher-order belief β representing subjective model fitness. An affective charge signal ϕ drives updates to β , rising when the model performs better than a prior baseline and falling when it performs worse. On this view, affect is metacognitive rather than mentalistic: it tracks model fitness rather than partner intentions or reward history alone.

This account has not, to our knowledge, been extended to relationship-specific social metacognition in multi-partner settings. The extension matters because model-fitness estimates are inherently relational. A person can carry a well-validated model of a longtime collaborator, a fragile model of a new acquaintance, and an actively revising model of someone who has recently behaved unexpectedly—and these confidence states should not be collapsed into a single shared precision budget. We therefore give each relationship its own β estimate and action-confidence signal, updated from that partner’s behavioral evidence alone.

We test the dynamics this produces in a repeated multi-partner trust game across binary and graded decision regimes, partner-choice conditions, abrupt betrayal, and profile-style parameter perturbations. The results characterize both the functional role and the temporal boundary conditions of partner-local social metacognition. The paper tests three linked claims: first, partner-local affective precision tracks model reliability rather than realized payoff; second, its behavioral effects arise from metacognitive action confidence rather than

improved partner-state beliefs; and third, the dynamics of precision gain and prior model fitness define temporally structured profiles of social trust revision.

2 Methods

A focal active-inference agent plays a repeated trust game with four partners whose types and stances are hidden and must be inferred from behavioral evidence. The agent maintains a separate generative model for each partner, allowing evidence and confidence to remain relationship-specific. Partner-local affective precision is layered on top of these models: each partner has its own model-fitness estimate, which modulates how confidently the agent acts on that partner’s inferred state.

2.1 Trust game

The trust game [4, 14] is a natural testbed because partner types are hidden and must be inferred, creating a genuine gap between social knowledge and actionable confidence. The policy space can also be varied: binary settings offer two actions, graded settings introduce several investment levels, and partner-choice regimes make approach and avoidance explicit by requiring the agent to select whom to engage.

Each partner has a latent *type* (cooperator, reciprocator, exploiter, or random) and a latent *stance* toward the agent (trusting, neutral, or hostile)—neither directly observable; both must be inferred from behavioral evidence. Episodes run for up to 200 rounds.

In the binary game both players choose to cooperate or defect. Table 1 gives the focal agent’s payoffs.

Table 1: Focal agent payoffs in the binary trust game.

	Partner cooperates Partner defects	
Agent cooperates	3	−1
Agent defects	5	1

The structure poses a social inference problem: defection is locally tempting, but cooperative partners are worth identifying and maintaining, so the agent must infer type and stance to act well over repeated interaction. In the graded regime, the binary choice is replaced by a discretized investment level; the agent commits a resource along a continuous scale rather than making a binary commitment. In the partner-choice regime, the agent selects which partner to engage before choosing an action each round.

2.2 Per-partner generative models

Given this environment, the focal agent maintains a separate discrete generative model $\mathcal{M}_k$ for each partner k , rather than a single pooled social model. This factorization keeps evidence about each partner independent and allows action confidence to vary across relationships. Each model encodes observation likelihoods, transition dynamics, preferences, priors, and policy priors using the standard *A/B/C/D/E* structure of discrete active inference. Payoff is represented as an observation modality rather than as an external reward cache, so preferences over outcomes remain part of the agent’s generative model. State and policy inference use the `pymdp` package for discrete active inference [12, 6].

2.3 Generative model specification

Each partner model contains three latent factors: partner type $s_k^{\text{type}} \in \{\text{cooperator, reciprocator, exploiter, raider}\}$, partner stance $s_k^{\text{stance}} \in \{\text{trusting, neutral, hostile}\}$, and a deterministic own-action bookkeeping factor s^{own} . The own-action factor is not a psychological hidden variable; it makes the focal agent’s executed action available to the payoff likelihood in a standard factorized POMDP representation. Observations are partner action $o_k^{\text{act}} \in \{\text{cooperate, defect}\}$ and focal payoff $o_k^{\text{pay}} \in \{-1, 1, 3, 5\}$.

The partner-action likelihood A_k^{act} is defined by type- and stance-conditioned cooperation probabilities. This table operationalizes predictive reliability: both reliably cooperative and reliably defecting partners can be predictable, even though their payoff consequences differ. Payoff enters as an observation modality $P(o_k^{\text{pay}} \mid s^{\text{own}}, s_k^{\text{type}}, s_k^{\text{stance}})$, with partner action marginalized through the cooperation likelihood. Thus payoff is deterministic in the environment but represented as a likelihood in the focal agent’s generative model. Stance transitions are action-dependent: cooperation tends to build trust, defection tends to damage trust, and recovery from hostility is slower than collapse. Full *A/B/C/D/E* matrices, cooperation probabilities, stance-transition matrices, priors, and payoff construction are provided in Appendix A.

2.4 Partner-local affective precision

Hesp et al. [13] treat policy precision as an inferred quantity rather than a fixed hyperparameter. The agent maintains a higher-order belief β over inverse policy precision, representing uncertainty about subjective model fitness: how reliably the generative model is supporting confident action. An affective charge signal ϕ drives updates to β , shifting the agent toward higher confidence when the model performs better than a prior baseline and toward lower confidence when it performs worse.

We extend this to the per-partner case by giving each partner its own β_k estimate, updated from that partner’s behavioral evidence alone. After interacting with partner k , the agent computes surprisal from the predicted probability of the observed partner response:

$$\epsilon_k = -\log P(o_k^{\text{act}} = \hat{o}_k^{\text{act}} \mid q(s_k^{\text{type}}, s_k^{\text{stance}})). \quad (1)$$

This is computed from partner *action* rather than payoff because action is the most direct test of type-by-stance model fit: a reliably defecting partner produces no action surprise even though the payoff is poor. We set the neutral fitness baseline at $\sigma_0^2 = (-\log 0.5)^2$, so a fifty-fifty binary prediction carries zero affective charge. Affective charge is then:

$$\phi_k = \alpha(\sigma_0^2 - \epsilon_k^2), \quad (2)$$

where α is a gain parameter. Positive ϕ_k means partner k 's model predicted better than the baseline; negative ϕ_k means it predicted worse. The categorical posterior $q(\beta_k)$ is updated by combining a slowly evolving persistence prior with a charge-based pseudo-likelihood that shifts mass toward low β after accurate prediction and toward high β after surprise; the full update rule is given in Appendix B.

The agent then uses the posterior mean to set policy precision:

$$\gamma_k = \gamma_{\text{base}} / \mathbb{E}_{q(\beta_k)}[\beta_k]. \quad (3)$$

Here, β_k represents inverse precision or model-fitness uncertainty: higher β_k means the model has been less reliable, producing a broader, less committed policy distribution; it is the inverse relationship that drives action sharpening. The mechanism thus translates accumulated evidence about predictive reliability into action confidence—independently for each social relationship.

The β_k posterior is therefore treated as an auxiliary precision tracker rather than as a hidden state within the generative model; the implications of this design choice are discussed in Section 4.

Cross-partner policy selection. In partner-choice regimes, policies specify both which partner to engage and how to act toward that partner. Each candidate policy branch inherits the precision of the partner it targets. Precision is applied to centered branch logits so that γ_k sharpens or flattens action selection within a partner branch without changing that branch's mean score in cross-partner comparison. This allows partner-local confidence to affect deployment while avoiding an artifact in which low precision makes uniformly poor partner branches spuriously attractive. Full details are provided in Appendix B.

Partner implementation. Reported experiments use one focal active-inference agent interacting with four environment-side partners. Partners maintain latent type and stance, sample actions from the type-by-stance cooperation probabilities, and undergo action-dependent stance transitions, but they do not perform variational inference or maintain their own affective precision estimates. This isolates the focal agent's partner-local precision mechanism before extending the model to reciprocal active-inference partners. Full implementation details are provided in Appendix A.

Simulation scale. Unless otherwise stated, simulations use 200-round episodes. Profile-program and mixed-volatility experiments use 20 seeds per condition,

while central behavioral comparisons use 30 seeds per condition. The abrupt-betrayal condition uses 120 rounds because the switch is scheduled explicitly and the main readouts are post-switch behavior. Full condition definitions, schedules, metrics, and seed counts are provided in Appendix C.

Code availability. Simulation code, analysis scripts, and experiment configurations will be released in a public repository upon acceptance.

3 Results

Simulations evaluate three linked claims about relationship-specific confidence. First, partner-local affective precision should track predictive reliability rather than realized value, and this signal should be weakened when evidence is pooled across partners. Second, its behavioral effects should arise through action sharpening rather than improved partner-state beliefs, with consequences for both policy entropy and partner engagement. Third, under social change, precision dynamics should reveal temporal boundary conditions and parameterized profiles of trust revision. The subsections below test these claims across reliability probes, shared- β ablations, deployment controls, partner-choice regimes, abrupt betrayal, and profile-style perturbations.

3.1 Precision tracks predictability, not partner value

The affective charge formula uses action surprisal rather than payoff, so β_k is sensitive to behavioral consistency rather than realized value. Once the agent has inferred a partner’s type and stance, a reliably defecting partner can generate low surprisal despite poor payoff, while an erratic partner can generate high surprisal despite occasionally favorable outcomes. The precision signal is therefore expected to track the quality of the agent’s predictive model, not the desirability of the partner’s behavior.

The 30-seed model-fitness confirmation provides the paper’s clearest reliability–value dissociation. Across partner-seed units, partner-local β_k shows a stronger absolute association between precision and action surprisal than between precision and realized payoff (0.949 versus 0.748). Because payoff, surprise, and exposure are correlated in partner-choice episodes, the stricter active-encounter partial readout is the more important test: after controlling encounter imbalance, the partner-local precision–surprise association remains strong (0.882), whereas the precision–payoff association falls to 0.130. A single shared β tracker preserves the ordering but with weaker relational specificity (partial associations 0.380 versus 0.049). Aggregate payoff moves in the opposite direction (partner-local: 483.5; shared: 517.6; no-affect: 542.1), which is precisely why this result is diagnostic: partner-local affective precision is not simply tracking reward or improving payoff. It supports the central mechanistic claim that partner-local precision tracks predictive model fitness more cleanly than realized partner value. Figure 1 visualizes this dissociation for partner-local versus shared precision tracking.

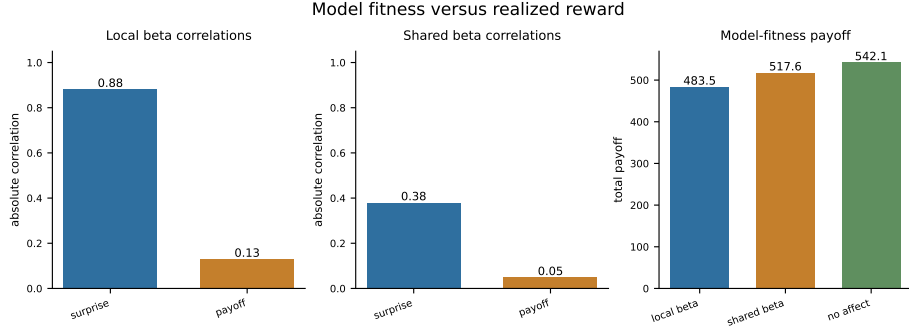


Fig. 1: Partner-local precision tracks action surprisal more strongly than realized payoff. The panel contrasts active-encounter partial associations between mean partner precision and surprise versus payoff for partner-local β_k and a shared β tracker, plus aggregate payoff. Partner-local tracking preserves the surprise-over-reward dissociation; pooling evidence across partners attenuates relational specificity.

3.2 Shared precision tests relational specificity

The locality probe above tests whether confidence remains tied to the partner whose evidence produced it. Pooling evidence across partners weakens this relational specificity: a shared β tracker preserves the coarse ordering between predictive surprise and realized payoff, but attenuates the partner-specific dissociation observed under partner-local precision. In the active-encounter partial readout, partner-local precision shows a stronger association with surprise than payoff, whereas shared precision compresses both associations toward weaker values.

This result should be interpreted as a mechanistic locality test rather than a definitive behavioral advantage for partner-local precision. The present shared- β comparison shows that pooling evidence blurs the model-fitness signal across relationships; it does not by itself establish that shared precision produces worse partner allocation in every mixed social environment. The mixed-volatility analyses below provide a related boundary condition: stronger precision dynamics can increase payoff while also increasing false-positive confidence reduction toward stable partners. A direct shared-versus-local mixed-volatility behavioral comparison would be the natural follow-up test, but the current claim is narrower: relationship-specific β_k preserves relational specificity in the affective precision signal.

3.3 Behavioral effects arise through action sharpening

A tracked-only ablation updates $q(\beta_k)$ normally but fixes $\gamma_k = \gamma_{\text{base}}$, allowing the precision estimate to be analyzed without letting it affect action selection.

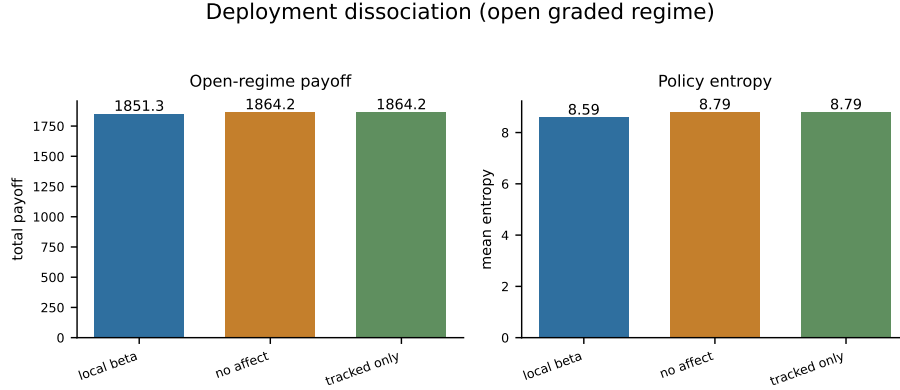


Fig. 2: Open graded regime: partner-local affective precision lowers policy entropy relative to no-affect and tracked-only controls without a corresponding payoff advantage. Tracked-only updates β_k but fixes γ_k ; its deployment readouts match the no-affect baseline, isolating the entropy effect to the $\beta_k \rightarrow \gamma_k$ action-confidence pathway.

In the open graded decision regime, full partner-local precision lowers policy entropy relative to both baselines (8.6 versus 8.8 for no-affect and tracked-only), while total payoff is flat-to-slightly lower (1851.3 versus 1864.2 for both baselines). Because tracked-only remains near the no-affect baseline while full precision modulation differs from both on entropy, the behavioral effect is localized to the $\beta_k \rightarrow \gamma_k$ deployment pathway rather than improved partner-state inference. Figure 2 summarizes the three-way contrast across affect, no-affect, and tracked-only variants.

3.4 Partner engagement reorganizes around reliability

In partner-choice regimes the precision signal shapes not only how the agent behaves toward a chosen partner but which partners it selects in the first place. Policy entropy under precision modulation is 4.66 versus 4.81 without it, and engagement shows a reliability-consistent shift. The partner-local agent selected the cooperator on 36.6% of rounds and the reciprocator on 31.6%, while allocating 13.8% and 18.0% of choices to the exploiter and random partners, respectively. The no-affect control selected the same partners at 34.8%, 31.8%, 16.2%, and 17.2%, respectively. Aggregate payoff is flat at this scale (377.3 versus 385.3), consistent with selection reorganization occurring before cumulative payoff differences have time to accumulate.

These allocation readouts support social reorganization rather than a definitive payoff advantage at this replication scale.

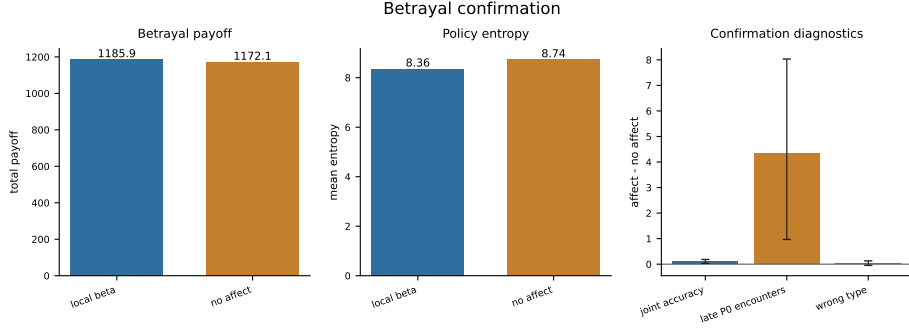


Fig. 3: Abrupt betrayal in the partner-choice regime (30 seeds). Partner-local affect lowers policy entropy relative to the no-affect control; the payoff difference is positive but small, and its bootstrap interval crosses zero. Joint partner-type accuracy is also higher under precision modulation, so the confirmed result is not a payoff–accuracy tradeoff but a deployment change with modest behavioral benefit.

3.5 Abrupt betrayal reveals the mechanism’s temporal structure

When a previously cooperative partner switches stance mid-episode, the agent enters the post-switch period with β_k reflecting a well-validated model that no longer applies. In the 30-seed abrupt-betrayal analysis, partner-local affective precision produces a small payoff advantage relative to the no-affect control (1185.9 versus 1172.1), but the paired bootstrap interval for the difference crosses zero (-25.2 to 53.2). The robust effect is instead the deployment signature: policy entropy is lower under precision modulation (8.36 versus 8.74; bootstrap interval for the difference -0.62 to -0.14), confirming that the action-confidence pathway is active.

Joint partner-type accuracy is higher under precision modulation (0.372 versus 0.266; bootstrap interval for the difference 0.034 to 0.185). The confirmed readout is therefore deployment change under abrupt social change: partner-local precision produces a robust entropy effect, higher joint partner-type accuracy, and an uncertain payoff advantage. The temporal-dependency interpretation captures when accumulated confidence helps or hurts depending on portfolio structure and reallocation speed, not whether affect generally improves reward.

This temporal dependence motivates the parameter analyses below: if accumulated confidence can help or hurt depending on the speed and structure of change, then precision gain and prior model fitness should define distinct profiles of social trust revision.

3.6 Precision dynamics as computational profiles of social trust calibration

The parameters governing β_k dynamics define a controlled profile program. Variation in precision gain α and the prior over model fitness produces qualitatively distinct behavioral profiles across social contexts. These profiles are computational analogues of empirically characterized variation in social trust calibration [3, 1, 5], not clinical classifications or fitted human profiles.

Profile-inspired computational perturbations (Appendix D) show that flattened precision dynamics can collapse action commitment without improving payoff: the low-gain variant compresses β_k range to 0.24 and raises entropy to 8.9, near the no-affect and tracked-only baselines, whereas expanded-dynamics variants widen β_k movement without monotonic payoff gains. This motivates parameter sweeps over gain and prior structure rather than a single high-versus-low trust axis.

Precision-gain axis. Sweeping α across $\{0.05, 0.1, 0.3, 0.5, 1.0, 2.0, 4.0, 8.0\}$ shows that β_k movement range increases monotonically with α in the betrayal regime. Low- α agents ($\alpha \leq 0.1$) show near-zero exploitation rates and Gini coefficients of 0.423 and 0.397, producing diffuse, slowly-updating engagement patterns. High- α agents ($\alpha \geq 4.0$) show Gini coefficients of 0.366 and 0.408 with recovery times of 28.8 and 13.0 rounds respectively—faster detection of behavioral change in some runs, but higher variance in recovery outcomes. Payoff does not rank α values monotonically across either regime (betrayal: range 1914.6–1986.2; open graded: 1906.3–1987.7), indicating that the behavioral consequence of precision-gain depends on task context rather than gain magnitude alone. Extended alpha-sweep diagnostics are in Appendix D.

Profile quadrants. Crossing prior belief (naive versus cautious) with α (low versus high) defines four behavioral profiles tested in the betrayal environment (20 seeds). Low- α profiles (naive-stubborn and avoidant-rigid) show substantially smaller β_k range (0.32–0.36) than high- α profiles (0.89–1.08), confirming that gain controls the magnitude of precision dynamics. Naive-stubborn shows the lowest Gini (0.304), reflecting maximally diffuse partner selection consistent with a slowly-updating confidence profile. Avoidant-rigid shows the lowest payoff (1889), as the cautious prior prevents confidence from building even toward reliably cooperative partners. The anxious-reactive profile shows the highest post-betrayal commitment errors toward the switched partner (post-betrayal P0 selection: 0.319; high investment rate: 0.121), reflecting rapid confidence accumulation that becomes a liability when a partner’s behavior changes. The default agent outperforms all profile variants on payoff (1991), occupying a calibrated region of the gain-prior space. Quadrant figures and supporting diagnostics are in Appendix D.

Forgiveness and trust repair. In the forgiveness paradigm, the betrayed partner reverts to cooperative/trusting behavior after round 120. The main pattern

separates reengagement from restored confidence. No-affect and cautious-low- α agents reengage most often after repair (selection rates 0.593 and 0.630), while default and naive-high- α agents reengage less (0.475 and 0.560). Payoff recovery is near baseline across all conditions (0.996–1.033), so the experiment does not support a simple "more affect means better forgiveness" claim. Low gain produces muted β_k dynamics and rapid reapproach in this task (cautious-low latency 1.5 rounds; naive-low 4.6), whereas high-gain profiles show larger confidence swings but not a higher payoff-recovery ceiling. Trust repair is therefore a profile-dependent dissociation: willingness to sample the reverted partner and restoration of model-fitness confidence can move separately.

Mixed-volatility discrimination. The most ecologically realistic test places four partners with heterogeneous volatility in the same episode: a stationary cooperator (P0), a stationary exploiter (P1), an abrupt-switch partner (P2), and a gradual-drift partner (P3). Under partner choice, the agent must maintain stable engagement with P0 while updating away from P2 and P3 after their behavioral changes.

Table 2: Mixed-volatility metrics by precision profile (20 seeds).

Variant	Payoff	Discrimination	P0 false positive	β range
default	1976	0.121	0.422	0.933
high- α	2054	−0.089	0.484	1.049
low- α	1927	0.213	0.359	0.182
no-affect	1934	—	0.273	—

The false-positive rate—unwarranted confidence reduction toward the stationary P0—increases with α : high- α (0.484), default (0.422), low- α (0.359), no-affect (0.273). The discrimination index does not simply improve with gain: low- α gives the strongest discrimination score (0.213), while high- α is negative (−0.089) despite achieving the highest payoff. Figure 4 shows why the result should not be read as a monotonic gain advantage. The mixed-volatility result therefore supports a narrower conclusion: high gain amplifies reactivity and false positives toward stable partners, but the payoff consequences depend on how that reactivity interacts with partner allocation.

Together, the profile experiments characterize affective precision dynamics not as a single behavioral tendency but as a family of profiles determined by gain and prior initial conditions. The key result is not a monotonic payoff ranking, but a dissociation between confidence dynamics, partner allocation, and payoff across social environments. Extended profile tables, alpha-sweep diagnostics, and forgiveness/trust-repair analyses are provided in Appendix D.

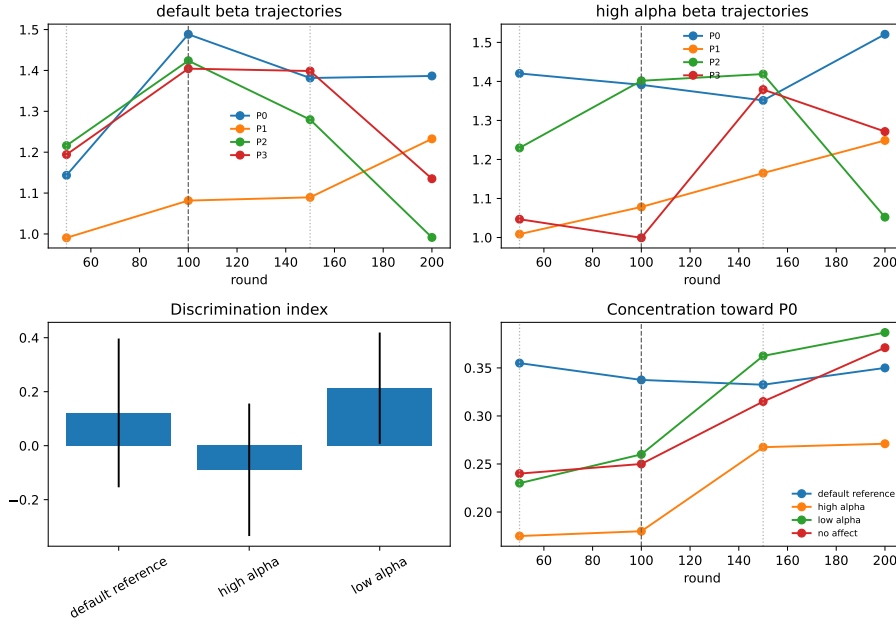


Fig. 4: Mixed-volatility partner-choice environment (20 seeds). Four partners differ in volatility: stationary cooperator (P0), stationary exploiter (P1), abrupt switch (P2), and gradual drift (P3). Higher precision-gain α can raise payoff while worsening discrimination and false-positive suppression toward P0; profiles are computational analogues, not clinical classifications.

4 Discussion

Social metacognition as the paper’s central object. The central contribution of this paper is to treat social metacognition as a distinct computational object—separate from partner-type inference and separate from theory-of-mind inference about what another agent knows or intends [10, 17]. Partner-local affective precision implements this object: an estimate not of what a partner will do, but of how well the focal agent’s current model of that partner justifies confident action. It evaluates the reliability of the focal agent’s own partner model and is therefore metacognitive rather than mentalistic.

The three problems are complementary rather than redundant. An agent can infer partner type accurately, maintain sophisticated ToM representations, and still be poorly calibrated about when to act confidently on those inferences—particularly when a partner is reliable in some contexts and unreliable in others, or when multiple relationships must be managed concurrently. Partner-local β_k addresses exactly this gap: action confidence tracks the recent track record of each partner model, independently of the model’s representational content. This complementarity is closest to ToM active-inference accounts of social cognition

[17], while factorized active-inference models provide a formal route for separating multiple latent components of strategic interaction [18]. The combination would permit an agent to infer what a partner is like, model what they know or intend, *and* maintain a running estimate of how much those models currently justify decisive action—a decomposition that may be necessary for robust performance in complex multi-partner environments.

Individual differences and the computational basis of social trust variation. The profile program in Section 3.6 does not claim that the four simulated profiles are validated clinical categories. Its contribution is more specific: it shows how two parameters, precision gain α and the prior over model fitness, can vary the *speed*, *amplitude*, and *asymmetry* of social confidence revision. This is the distinction a scalar trust account misses. Two agents can have similar average trust while differing sharply in whether they update quickly after negative evidence, recover after renewed cooperation, or overreact to noise from otherwise stable partners.

This framing connects the simulations to empirical literatures on social learning, social anxiety, psychosis, and attachment without reducing those constructs to model labels [3, 1, 5]. These paradigms matter because they separate dimensions that scalar trust models collapse: reengagement versus restored confidence, sensitivity to change versus false positives, and payoff improvement versus accurate discrimination.

Betrayal reveals a temporal dependency in confidence deployment. The betrayal result makes this temporal dependency concrete. In the 30-seed abrupt-betrayal analysis, partner-local affective precision lowered policy entropy and increased joint partner-type accuracy, while producing only a small payoff advantage whose bootstrap interval crossed zero. The confirmed readout is deployment change under abrupt social change: accumulated relationship-specific confidence reshapes how the agent acts, with benefits depending on whether the surrounding partner portfolio allows rapid reallocation away from the changed partner.

The implication is structural. A confidence mechanism built from evidence accumulation has a temporal dependency: its benefit under abrupt change depends on whether accumulated confidence can be productively redirected, whether alternative partners exist, and whether reallocation occurs faster than stale confidence can keep action attached to the changed partner. This suggests a natural extension: a volatility or change-detection channel that discounts accumulated confidence when prediction-error structure changes. This is related to the problem addressed by hierarchical volatility models [15, 3], but it appears here at the level of action confidence rather than learning rate.

Limitations. Four design features bound the current claims. First, β_k is maintained as an auxiliary precision tracker rather than as a hidden state in the generative model; in the full Hesp et al. hierarchy it is a proper inferred hidden state, allowing the agent to form prior expectations over its own future affective states and to plan around anticipated changes in partner reliability. Making β_k a variational variable would enable richer prospective regulation of action

confidence. Second, partners are parameterized environment-side policies rather than reciprocal active-inference agents. This isolates the focal mechanism, but it means the present simulations do not address the reciprocal case in which both agents update their beliefs, confidence, and policies simultaneously. Third, the profile taxonomy in Section 3.6 is predictive rather than validated against human data: the experiments test whether different α and prior values produce qualitatively distinct behavioral profiles, and these simulations connect those targets to human individual differences in social trust. A stronger claim—that individual differences in human social trust calibration arise specifically from variation in an affective precision mechanism with this structure—would require fitting the model to human behavioral data and establishing that parameter estimates predict theoretically relevant outcomes. The profile predictions are offered as a scaffold for such empirical work, not as an established correspondence. Fourth, several behavioral effects should be interpreted cautiously because they are modest at the current simulation scale. The betrayal payoff advantage is small and its paired bootstrap interval crosses zero, and the profile-style experiments use 20 seeds per condition. The strongest supported claims are therefore the mechanistic dissociations that replicate across controls—precision tracking surprise more strongly than payoff, entropy reduction through the $\beta_k \rightarrow \gamma_k$ pathway, and attenuation under shared β —rather than every payoff ordering observed in the simulations.

Future directions. The most important near-term extension is empirical: fitting the affective precision model to human behavioral data from iterated trust game paradigms with individual-difference measures. The profile analyses in Section 3.6 generate quantitative predictions about exploitation sensitivity, betrayal recovery time, trust-revision asymmetry, and mixed-volatility false positives that are measurable from behavioral time series without neuroimaging. The α and prior parameters could be estimated per participant and tested against personality measures, attachment style, and clinical scores to assess whether the computational profiles correspond to real psychological variation. A second extension is ecological and reciprocal: replacing environment-side partner policies with active-inference partners that maintain their own beliefs, precision estimates, and policies, then embedding these agents in settings with communication, reputation, and coalition structure. Combining this signal with ToM-style belief inference [17] would allow future work to study whether metacognitive action-confidence signals and mentalistic partner-state models are functionally coupled or dissociable in social decision-making.

5 Conclusion

Predicting what a partner will do, modeling what they know or intend, and deciding how confidently to act on one’s current partner model are separable problems. Partner-local affective precision addresses the third as a relationship-specific metacognitive confidence signal. The simulations show that it tracks

predictive reliability more than realized value, affects action deployment through policy precision rather than improved partner-state inference, and creates temporally structured trust-revision profiles under social change. By giving each social relationship its own model-fitness estimate, the mechanism allows action confidence to be calibrated independently across a social portfolio. Precision gain and priors over model fitness define a family of computational trust profiles for individual-difference research, positioning the framework as a mechanistically grounded account of social affect.

References

1. Adams, R.A., Stephan, K.E., Brown, H.R., Frith, C.D., Friston, K.J.: The computational anatomy of psychosis. *Frontiers in Psychiatry* **4**, 47 (2013). <https://doi.org/10.3389/fpsyt.2013.00047>
2. Barrett, L.F.: Solving the emotion paradox: Categorization and the experience of emotion. *Personality and Social Psychology Review* **10**(1), 20–46 (2006). https://doi.org/10.1207/s15327957pspr1001_2
3. Behrens, T.E.J., Hunt, L.T., Woolrich, M.W., Rushworth, M.F.S.: Associative learning of social value. *Nature* **456**, 245–249 (2008). <https://doi.org/10.1038/nature07538>
4. Berg, J., Dickhaut, J., McCabe, K.: Trust, reciprocity, and social history. *Games and Economic Behavior* **10**(1), 122–142 (1995). <https://doi.org/10.1006/game.1995.1027>
5. Cittern, D., Nolte, T., Friston, K., Edalat, A.: Intrinsic and extrinsic motivators of attachment under active inference. *PLOS ONE* **13**(4), e0193955 (2018). <https://doi.org/10.1371/journal.pone.0193955>
6. Da Costa, L., Parr, T., Sajid, N., Veselic, S., Neacsu, V., Friston, K.J.: Active inference on discrete state-spaces: A synthesis. *Journal of Mathematical Psychology* **99**, 102447 (2020). <https://doi.org/10.1016/j.jmp.2020.102447>
7. Friston, K.: The free-energy principle: a unified brain theory? *Nature Reviews Neuroscience* **11**, 127–138 (2010). <https://doi.org/10.1038/nrn2787>
8. Friston, K., FitzGerald, T., Rigoli, F., Schwartenbeck, P., Pezzulo, G.: Active inference: A process theory. *Neural Computation* **29**(1), 1–49 (2017). <https://doi.org/10.1162/NECO'a'00912>
9. Friston, K., Rigoli, F., Ognibene, D., Mathys, C., Fitzgerald, T., Pezzulo, G.: Active inference and epistemic value. *Cognitive Neuroscience* **6**(4), 187–214 (2015). <https://doi.org/10.1080/17588928.2015.1020053>
10. Frith, C.D., Frith, U.: Mechanisms of social cognition. *Annual Review of Psychology* **63**, 287–313 (2012). <https://doi.org/10.1146/annurev-psych-120710-100449>
11. Hampton, A.N., Bossaerts, P., O'Doherty, J.P.: Neural correlates of mentalizing-related computations during strategic interactions in humans. *Proceedings of the National Academy of Sciences* **105**(18), 6741–6746 (2008). <https://doi.org/10.1073/pnas.0711099105>
12. Heins, C., Millidge, B., Demekas, D., Klein, B., Friston, K., Couzin, I.D., Tschantz, A.: pymdp: A python library for active inference in discrete state spaces. *Journal of Open Source Software* **7**(73), 4098 (2022). <https://doi.org/10.21105/joss.04098>
13. Hesp, C., Smith, R., Parr, T., Allen, M., Friston, K.J., Ramstead, M.J.D.: Deeply felt affect: The emergence of valence in deep active inference. *Neural Computation* **33**(2), 398–446 (2021). <https://doi.org/10.1162/neco'a'01341>

14. King-Casas, B., Tomlin, D., Anen, C., Camerer, C.F., Quartz, S.R., Montague, P.R.: Getting to know you: reputation and trust in a two-person economic exchange. *Science* **308**(5718), 78–83 (2005). <https://doi.org/10.1126/science.1108062>
15. Mathys, C., Daunizeau, J., Friston, K.J., Stephan, K.E.: A Bayesian foundation for individual learning under uncertainty. *Frontiers in Human Neuroscience* **5**, 39 (2011). <https://doi.org/10.3389/fnhum.2011.00039>
16. Parr, T., Pezzulo, G., Friston, K.J.: *Active Inference: The Free Energy Principle in Mind, Brain, and Behavior*. MIT Press (2022)
17. Pitliya, R.J., Çatal, O., Van de Maele, T., Pezzato, C., Verbelen, T.: Theory of mind using active inference: A framework for multi-agent cooperation (2025). <https://doi.org/10.48550/arXiv.2508.00401>
18. Ruiz-Serra, J., Sweeney, P., Harré, M.S.: Factorised active inference for strategic multi-agent interactions. In: *Proceedings of the 24th International Conference on Autonomous Agents and Multiagent Systems*. pp. 1793–1802. International Foundation for Autonomous Agents and Multiagent Systems (2025). <https://doi.org/10.5555/3709347.3743815>

Appendix

A Full POMDP Specification

Scope. This appendix provides supporting technical detail for the corresponding main-text sections. Essential claims and results remain in the main paper.

A.1 Hidden state factors and observations

The focal agent’s generative model for each partner k maintains three latent factors:

$$s_k^{\text{type}} \in \{\text{cooperator, reciprocator, exploiter, random}\} \quad (\text{latent behavioral type}) \quad (4)$$

$$s_k^{\text{stance}} \in \{\text{trusting, neutral, hostile}\} \quad (\text{latent stance}) \quad (5)$$

$$s^{\text{own}} \in \mathcal{A} \quad (\text{agent’s executed social action}) \quad (6)$$

The own-action state is *deterministic bookkeeping*: the agent always knows its own action, and this factor makes that knowledge available to the observation model. It is not a psychological hidden variable. In the binary task, the joint hidden state per partner spans $4 \times 3 \times 2 = 24$ possibilities; graded regimes replace the binary action factor with the configured investment levels.

Observations are partner action and payoff:

$$o_k^{\text{act}} \in \{\text{cooperate, defect}\} \quad (\text{partner response}) \quad (7)$$

$$o_k^{\text{pay}} \in \{-1, 1, 3, 5\} \quad (\text{focal agent’s realized payoff}) \quad (8)$$

A.2 Partner-action likelihood

The core of the generative model is the *partner cooperation likelihood table*, which defines the probability the partner cooperates given type and stance:

Table 3: Cooperation likelihood $P(\text{cooperate} \mid s^{\text{type}}, s^{\text{stance}})$.

Partner type	Trusting	Neutral	Hostile
Cooperator	0.95	0.80	0.55
Reciprocator	0.90	0.70	0.30
Exploiter	0.70	0.35	0.10
Random	0.60	0.50	0.35

This table is the foundation of the likelihood modality A_k^{act} . Entries define

$$P(o_k^{\text{act}} = \text{cooperate} \mid s_k^{\text{type}}, s_k^{\text{stance}}), \quad (9)$$

with $P(o_k^{\text{act}} = \text{defect} \mid \cdot) = 1 - P(o_k^{\text{act}} = \text{cooperate} \mid \cdot)$. The modality is uniform over s^{own} : partner action depends only on inferred type and stance. This operationalizes what the agent considers a predictable partner—a cooperative reciprocator in a trusting stance has high predicted cooperation; an exploiter in a hostile stance has low predicted cooperation.

A.3 Payoff likelihood

Payoff enters as a second observation modality $A_k^{\text{pay}} = P(o_k^{\text{pay}} \mid s^{\text{own}}, s_k^{\text{type}}, s_k^{\text{stance}})$ rather than as external reward. The environment payoff table is deterministic (Table 1 in the main text); in the generative model, partner action is marginalized through the cooperation likelihood:

$$P(o_k^{\text{pay}} = v \mid s^{\text{own}}, s_k^{\text{type}}, s_k^{\text{stance}}) = \sum_{a^{\text{partner}} \in \{C, D\}} \mathbb{I}[\pi(s^{\text{own}}, a^{\text{partner}}) = v] P(o_k^{\text{act}} = a^{\text{partner}} \mid s_k^{\text{type}}, s_k^{\text{stance}}), \quad (10)$$

where $\pi(\cdot, \cdot)$ maps joint actions to focal payoffs $\{-1, 1, 3, 5\}$ and $P(o_k^{\text{act}} \mid \cdot)$ comes from Eq. (9). This preserves the Prisoner’s Dilemma incentive structure at horizon 1 while allowing trust-building dynamics to emerge over longer horizons.

A.4 Transition dynamics

Type drift (B_k^{type}). Partner type is uncontrollable and drifts slowly:

$$P(s_k^{\text{type}'} = j \mid s_k^{\text{type}} = i) = \begin{cases} 1 - p_{\text{switch}} & j = i, \\ p_{\text{switch}} / (K_{\text{type}} - 1) & j \neq i, \end{cases} \quad (11)$$

with $K_{\text{type}} = 4$ and default $p_{\text{switch}} = 0.05$. Scheduled betrayal and mixed-volatility scenarios set $p_{\text{switch}} = 0$.

Stance dynamics (B_k^{stance}). Stance transitions are action-dependent on the focal agent’s social action a :

$$P(s_k^{\text{stance}'} \mid s_k^{\text{stance}}, a), \quad (12)$$

as specified below. Cooperation tends to build trust; defection damages trust; recovery from hostility is slower than collapse.

Table 4: Stance transition dynamics $P(s'^{\text{stance}} | s^{\text{stance}}, a)$ conditional on focal agent action.

Agent cooperates			
To \ From	Trusting	Neutral	Hostile
Trusting	0.90	0.30	0.05
Neutral	0.10	0.60	0.35
Hostile	0.00	0.10	0.60
Agent defects			
To \ From	Trusting	Neutral	Hostile
Trusting	0.10	0.05	0.02
Neutral	0.50	0.35	0.18
Hostile	0.40	0.60	0.80

These dynamics instantiate the asymmetry of trust: cooperation gradually builds trust (neutral $\rightarrow$ trusting in ~ 3 –4 steps), while defection rapidly damages trust (trusting $\rightarrow$ hostile in ~ 2 steps); recovery from hostility is slow (~ 4 –5 steps to neutral).

Own-action bookkeeping (B^{own}). The own-action factor updates deterministically from the executed social action:

$$P(s^{\text{own}'} = a' | s^{\text{own}}, a) = \mathbb{I}[a' = a]. \quad (13)$$

This factor is not a psychological hidden variable; it makes the agent’s executed action available to the payoff likelihood in standard factorized form.

A.5 Preferences, priors, and policy priors

Observation preferences (C_k). There is no direct preference over partner actions:

$$C_k^{\text{act}} = [0, 0] \quad \text{over } \{\text{cooperate, defect}\}. \quad (14)$$

Payoff preferences ascend over $\{-1, 1, 3, 5\}$:

$$C_k^{\text{pay}} = \log_softmax\left(\frac{[-1, 1, 3, 5]}{\tau}\right), \quad (15)$$

where τ is a temperature parameter controlling preference sharpness. Instrumental motivation therefore enters through payoff observations and multi-step planning rather than direct action preferences.

Initial-state priors (D_k). Type, stance, and own-action priors are

$$D_k^{\text{type}} = \left[\frac{1}{4}, \frac{1}{4}, \frac{1}{4}, \frac{1}{4}\right], \quad (16)$$

$$D_k^{\text{stance}} = [0.2, 0.6, 0.2], \quad (17)$$

$$D^{\text{own}} = [0.5, 0.5] \quad (\text{binary task; uniform over graded levels in graded regimes}). \quad (18)$$

Policy priors (E_k). Policy priors are uniform over admissible action sequences at the configured planning horizon (no habitual bias toward cooperation or defection). For horizon 1 in the binary task, $E_k = [0.5, 0.5]$ over {cooperate, defect}; longer horizons enumerate all length- τ sequences with equal prior mass.

Partner selection across k is handled outside the per-partner POMDP by evaluating policy branches for each partner with precision-adjusted logits before executing the chosen action (Appendix B).

A.6 Partner implementation

Reported experiments use one focal active-inference agent with per-partner generative models as described above. Partners are parameterized environment-side policies, not reciprocal active-inference agents. Each partner maintains a latent type and stance, follows the type-by-stance cooperation probabilities in Table 3, and undergoes action-dependent stance transitions in Table 4. Partners do not run variational inference, maintain their own policy precision γ_k , or update affective estimates β_k . This design isolates the focal agent’s precision-tracking mechanism before adding reciprocal belief and policy updating on the partner side.

B Affective Precision Update and Policy Selection

Scope. This appendix provides supporting technical detail for the corresponding main-text sections. Essential claims and results remain in the main paper.

B.1 Categorical beta support

Each partner maintains a categorical posterior $q(\beta_k)$ over discrete levels $\{0.5, 0.67, 1.0, 1.5, 2.0\}$, updated each round from affective charge ϕ_k . The posterior is categorical rather than continuous; this keeps the inverse- β convention identifiable while preserving a compact implementation of the precision dynamics.

B.2 Persistence prior

The update proceeds in three steps. First, a persistence prior smooths the previous posterior via a tridiagonal transition matrix:

$$p(\beta_k) \leftarrow T \cdot q_{\text{prev}}(\beta_k), \quad (19)$$

where T is tridiagonal with persistence parameter 0.8 and reflecting boundaries, encoding the prior expectation that precision evolves slowly.

B.3 Charge-based pseudo-likelihood

Second, a pseudo-likelihood is constructed directly from affective charge, mapping positive charge to high precision (low β) and negative charge to low precision (high β):

$$\log \ell(\beta_\ell) = \phi_k \cdot \frac{1}{\beta_\ell}, \quad (20)$$

where β_ℓ ranges over the five support points. This is a hand-designed update rule (not derived from the POMDP generative model) chosen to satisfy monotonicity: strong positive charge reinforces low β , strong negative charge reinforces high β , and the effect scales with effective precision at each level.

B.4 Posterior update and policy precision

Third, the posterior is normalized:

$$q(\beta_k) \propto \exp(\log \ell(\beta_\ell)) \cdot p(\beta_k). \quad (21)$$

The agent then uses the posterior mean to set policy precision:

$$\gamma_k = \gamma_{\text{base}} / \mathbb{E}_{q(\beta_k)}[\beta_k]. \quad (22)$$

The β_k posterior is maintained as an external auxiliary tracker rather than a proper hidden state in the generative model, which simplifies implementation at the cost of losing the agent’s ability to form prior expectations over its own future affective states (discussed in Section 4).

B.5 Centered cross-partner policy logits

In partner-choice regimes, precision must sharpen policies within a partner branch without creating artifacts when branches are compared across partners. Let $u_{k,\pi}$ denote the pre-precision policy logit for policy π toward partner k , and let $\bar{u}_k$ be that partner branch’s mean logit. Precision is applied to centered logits:

$$\tilde{u}_{k,\pi} = \bar{u}_k + \gamma_k (u_{k,\pi} - \bar{u}_k). \quad (23)$$

The agent then samples or selects policies from the combined set of $\tilde{u}_{k,\pi}$ values. Centering preserves the mean score of each partner branch while allowing γ_k to sharpen or flatten the within-partner policy distribution. This prevents low-precision scaling from moving a uniformly poor branch toward zero and making it spuriously attractive in cross-partner comparison.

B.6 Ablation variants

Reported comparisons use four affect-runtime variants. The *no-affect* control disables the beta tracker and fixes policy precision at γ_{base} . The *tracked-only* ablation updates $q(\beta_k)$ normally but fixes $\gamma_k = \gamma_{\text{base}}$, allowing the precision

estimate to be analyzed without letting it affect action selection. The default *partner-local precision* variant applies the full update and sets γ_k from each partner’s posterior mean. The *shared- β* ablation maintains a single pooled beta state updated from partner evidence while keeping partner-local POMDP beliefs intact; all partner branches then inherit precision from the shared expected β .

C Simulation Protocols and Metrics

Scope. This appendix provides supporting technical detail for the corresponding main-text sections. Essential claims and results remain in the main paper.

C.1 Simulation conditions

All reported experiments use one focal active-inference agent with four partners ($K = 4$), per-partner generative models, and environment-side partner policies (Appendix A). Conditions differ by payoff regime, assignment mode, and scheduled partner changes.

Binary random-assignment baseline. In the binary trust game (Table 1 in the main text), partners are assigned randomly each round and the agent chooses cooperate or defect. Stochastic partner-type drift uses $p_{\text{switch}} = 0.05$ unless disabled for scheduled-switch scenarios.

Open graded decision regime. The graded regime replaces the binary action with a discretized investment level while keeping the same latent type-stance structure. This is the primary deployment-dissociation setting (Section 3.3): affect, no-affect, and tracked-only variants are compared on policy entropy and cumulative payoff.

Partner-choice regime. The agent first selects a partner, then selects an action toward that partner each round. Partner selection is where engagement reorganization and social-allocation readouts are defined (Section 3.4).

Focal-switch locality probe. A partner-choice graded environment with type volatility disabled ($p_{\text{switch}} = 0$) and a single scheduled stance switch on partner P0 tests whether local versus shared β tracking preserves the surprise-over-reward dissociation (Section 3.1).

Abrupt betrayal. A partner-choice graded environment with scheduled stance betrayal on a previously cooperative partner tests post-switch reallocation, entropy, payoff, and joint accuracy (Section 3.5).

Precision-dynamics perturbations. Profile-inspired computational variants (low-gain, high-gain, and cautious-prior profiles) perturb β_k amplitude and stability in the graded decision regime without claiming clinical validity (Appendix D).

Profile program (Exp A–D). Parameter sweeps and factorial designs vary precision gain α and prior over model fitness to define computational trust-calibration profiles in open graded, betrayal, partner-choice, forgiveness, and mixed-volatility environments (Section 3.6).

C.2 Seed counts and episode lengths

Unless otherwise stated in the main text, episode length is 200 rounds. The abrupt-betrayal condition uses 120 rounds because post-switch readouts dominate interpretation. Profile-program and mixed-volatility tables use 20 seeds per condition. Central behavioral comparisons use 30 seeds per condition. Exploratory diagnostic runs used three seeds per variant and were used only for implementation validation; all main-text claims are based on the confirmation-scale or profile-scale runs reported in the relevant sections.

Representative configured scales:

Table 5: Representative episode lengths and replication counts by experiment family.

Experiment family	Setting	Rounds	Seeds
H0/H2 open graded deployment	graded, random or open choice	200	30
H1 model fitness	graded / reliability probes	200	30
H3 focal-switch locality	partner choice, single switch	100	30
H4 partner choice	partner choice	200	30
H5 abrupt betrayal	partner choice, scheduled betrayal	120	30
H6 perturbation dynamics	graded perturbations	200	30
Exp A–D profiles	sweeps / factorial / forgiveness / mixed volatility	200	20

Supplementary materials. The supplementary repository contains the simulation environments, affect-runtime variants, experiment configuration files, analysis scripts, and figure-generation code used for the reported experiments.

C.3 Betrayal schedule

The canonical abrupt-betrayal protocol (H5 `betrayal_choice`) uses graded pay-offs, agent-chosen partners, four partners with fixed initial types and stances, and no stochastic type switching ($p_{\text{switch}} = 0$). Partner P0 begins as a cooperator in a trusting stance. At round 31, P0 undergoes a scheduled stance switch to hostile while other partners retain their initial configurations. Primary readouts are post-switch cumulative payoff, policy entropy, partner reallocation, and joint partner-type accuracy over the remaining episode.

The profile-program betrayal environment used in Exp A and Exp B applies the same single-switch structure within 200-round episodes for recovery-time and quadrant metrics. Exp C (forgiveness) extends this schedule: rounds 1–80 cooperative baseline, rounds 81–120 betrayal on P0, rounds 121–200 reversion to cooperative/trusting on P0.

C.4 Mixed-volatility schedule

The mixed-volatility environment (Exp D; Section 3.6) places four partners with heterogeneous volatility in the same partner-choice episode over 200 rounds:

- **P0:** stationary cooperator (no scheduled change).
- **P1:** stationary exploiter (no scheduled change).
- **P2:** abrupt shift—cooperative/trusting pre-switch, hostile/exploiter from the scheduled switch onward (round 101 in the implemented scenario).
- **P3:** gradual drift—stance and type transitions spread across mid-episode rounds (neutral stance from round 51; hostile/exploiter shift from round 151 in the implemented scenario).

Stochastic type switching is disabled ($p_{\text{switch}} = 0$). Conditions compare default, low- α , high- α , and no-affect variants at 20 seeds each.

C.5 Partner-choice allocation metrics

Partner-choice readouts quantify how precision reshapes *who* the agent engages, not only how it acts toward a fixed partner.

Selection share. Per-partner selection rate is the fraction of rounds on which each partner is chosen. Partner-choice reallocation is summarized qualitatively in Section 3.4.

Selection concentration. The Gini coefficient of the partner-selection distribution summarizes how concentrated engagement is across the social portfolio (used in profile Exp A and Exp B).

Post-betrayal allocation. Post-switch windows report reallocation away from the switched partner and toward alternatives, including post-betrayal P0 selection and high-investment rates in profile analyses.

Policy entropy under choice. Mean q_π entropy in partner-choice regimes captures how sharply the agent commits to partner-action policies (reported comparatively for affect versus no-affect controls).

C.6 Entropy, payoff, and accuracy readouts

Total payoff. Cumulative focal-agent payoff over the episode (or over defined post-switch windows for betrayal analyses).

Policy entropy. Mean q_π entropy (‘q-pi-entropy’ in logged outputs) measures action-commitment sharpness; lower entropy indicates sharper policy selection under precision modulation.

Joint partner-type accuracy. Mean joint correctness of inferred partner type (‘mean_joint_accuracy’) summarizes epistemic state inference; it is reported separately from payoff because precision modulation can change deployment and accuracy independently of cumulative reward.

β_k range. Per-partner or episode-aggregated range of β_k support summarizes precision-dynamics amplitude in profile and perturbation tables.

Model-fitness dissociation. Partner-local correlations between mean precision and action surprisal versus realized payoff test whether β_k tracks predictive reliability rather than partner value (Section 3.1).

Mixed-volatility discrimination metrics. The discrimination index is the Pearson correlation between per-partner β_k trajectory and per-partner behavioral stability (higher indicates better separation of stable versus shifting partners). The P0 false-positive rate is the rolling rate at which engagement with the stationary cooperator P0 falls more than 15% below its early baseline despite P0 never changing.

Betrayal timing metrics. Recovery time and related latency readouts measure rounds after a switch until selection or payoff returns toward baseline thresholds (profile Exp A/B and betrayal analyses).

C.7 Replication summary

Central behavioral comparisons in Section 3 use 30 seeds per condition. Profile-program and mixed-volatility analyses use 20 seeds per condition. The model-fitness dissociation (Section 3.1) uses active-encounter partial correlations to compare precision-surprise against precision-payoff associations, with shared β included as the relational-specificity ablation.

D Extended Phenotype and Robustness Results

Scope. This appendix provides supporting technical detail for the corresponding main-text sections. Essential claims and results remain in the main paper.

D.1 Precision-dynamics perturbations

To connect the basic mechanism to the phenotype program, we perturb the amplitude and stability of β_k dynamics using phenotype-inspired computational variants. These variants are not diagnostic models; they are controlled ways of asking how flattened, expanded, or unstable precision dynamics alter action commitment.

Table 6: Precision-dynamics perturbations in the graded decision regime. Variants are computational profiles, not clinical diagnoses.

Variant	β_k range	Total payoff	Policy entropy
affect	1.32	1851.3	8.6
low-gain	0.24	1840.5	8.9
high-gain	1.48	1859.0	8.1
cautious-prior	1.48	1877.7	8.4

Signal flattening produces the clearest behavioral consequence. The low-gain variant compresses β_k range to 0.24 and raises entropy to 8.9, near the no-affect and tracked-only baselines. Expanded-dynamics variants produce wider β_k movement, but payoff does not follow range monotonically. This motivates the parameter sweeps below: social confidence differences are better captured by gain and prior structure than by a single high-versus-low trust axis.

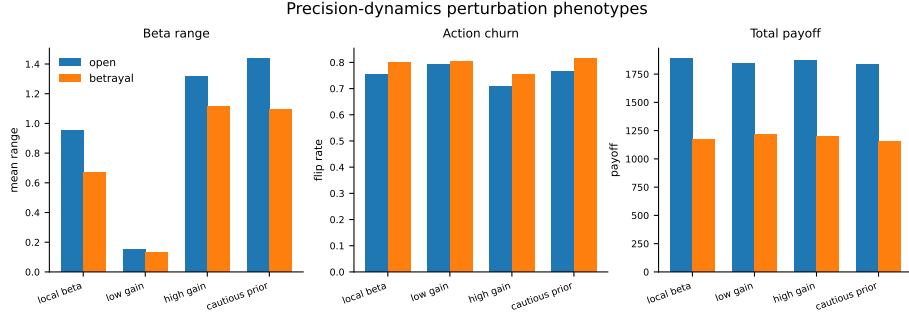


Fig. 5: Precision-dynamics perturbations in the graded decision regime. Variants are computational profiles, not clinical diagnoses or classifications.

D.2 Alpha sweep

Sweeping α across $\{0.05, 0.1, 0.3, 0.5, 1.0, 2.0, 4.0, 8.0\}$ tests whether precision gain controls reactive social confidence across metrics. β_k movement range increases monotonically with α in the betrayal regime:

Table 7: β_k range by precision-gain α (betrayal regime, 20 seeds).

α	Mean β_k range
0.05	0.147
0.1	0.182
0.3	0.326
0.5	0.416
1.0	0.580
2.0	0.862
4.0	1.051
8.0	1.218

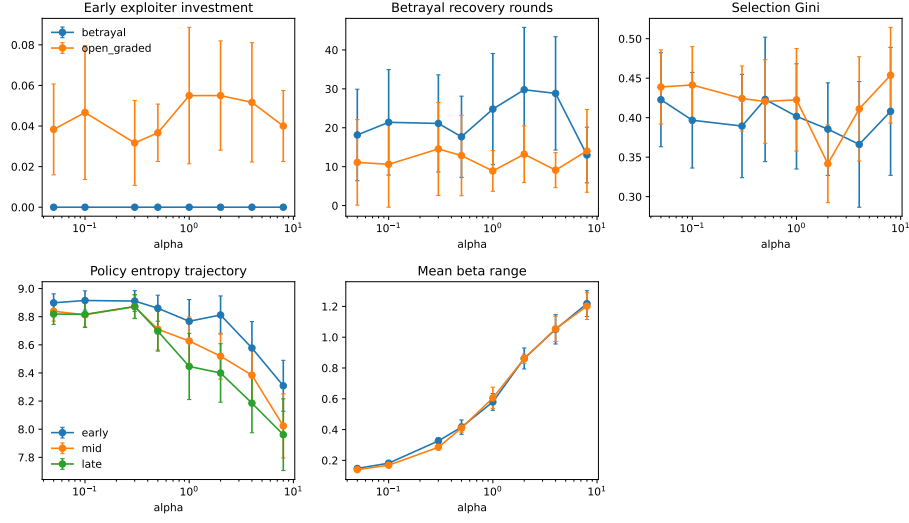


Fig. 6: Precision-gain sweep across open graded and betrayal environments (20 seeds). Higher α expands β_k range and reshapes engagement readouts without monotonic payoff ranking.

D.3 Phenotype quadrants

Crossing prior belief (naive versus cautious) with α (low versus high) defines four behavioral profiles tested in the betrayal environment (20 seeds). Table 8 gives key metrics.

Table 8: Behavioral metrics by phenotype (betrayal environment, 20 seeds). Default agent shown for reference.

Profile	Payoff	Recovery	Gini	β range	Trust asymmetry
Anxious-reactive	1932	26.5	0.338	1.084	0.98
Hypervigilant	1951	20.1	0.423	0.892	1.65
Naive-stubborn	1894	21.3	0.304	0.364	1.19
Avoidant-rigid	1889	14.9	0.357	0.322	1.21
Default	1991	23.0	0.423	0.893	1.68

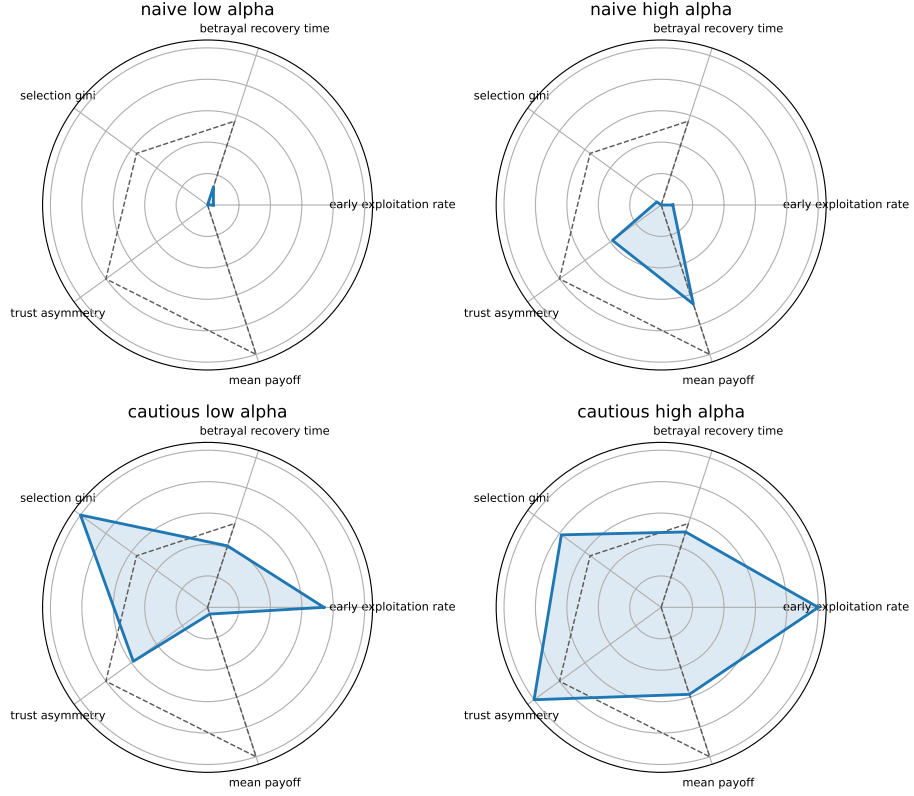


Fig. 7: Phenotype quadrant profiles from the prior-by- α factorial (betrayal environment, 20 seeds). Profiles are computational analogues, not clinical classifications.

D.4 Forgiveness and trust repair

The forgiveness paradigm uses a 200-round partner-choice episode: cooperative baseline (rounds 1–80), betrayal on P0 (rounds 81–120), then reversion to cooperative/trusting behavior (rounds 121–200). Readouts include reengagement latency, post-return selection, payoff recovery, and β_k recovery trajectories for the reverted partner.

Table 9: Forgiveness and trust-repair metrics by precision profile (20 seeds). Reengagement is the post-repair P0 selection rate; payoff recovery compares late post-repair payoff with the late pre-betrayal baseline.

Variant	Reengagement	Payoff recovery	Latency	β_k range
cautious-high- α	0.518	0.996	5.5	0.971
cautious-low- α	0.630	1.014	1.5	0.287
default	0.475	1.019	3.0	1.012
naive-high- α	0.560	1.005	2.2	1.100
naive-low- α	0.527	1.004	4.6	0.325
no-affect	0.593	1.033	1.7	—

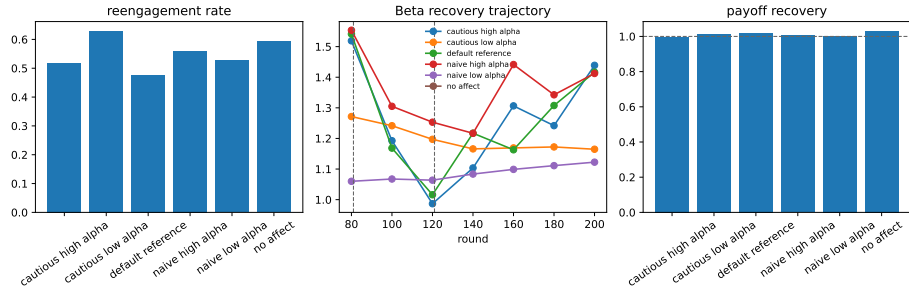


Fig. 8: Forgiveness/trust-repair diagnostic from the 20-seed phenotype program. Reengagement and payoff recovery do not collapse into a single monotonic gain effect: no-affect and cautious-low- α profiles reengage most often, while payoff recovery remains near baseline across profiles.

D.5 Mixed-volatility extended diagnostics

The primary mixed-volatility table and figure remain in Section 3.6. Extended Exp D diagnostics include per-partner β_k trajectories, rolling P0 selection concentration, and pairwise stability–precision correlations used to compute the discrimination index and P0 false-positive rate reported in the main text.